



The role of green shared vision in building organizational citizenship behavior for sustainability: A serial multiple mediation model

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ABSTRACT

This research attempts to develop and test a research framework to explore how green shared vision (GSV) impacts employee organizational citizenship behavior for the environment (OCBE) and the single and serial multiple mediation impacts of environmental corporate social responsibility (ECSR) and green work engagement (GWE). We garnered data from hotel employees in Seoul. Using structural equation modeling with the user-defined estimands function reveals that GSV has a significantly positive effect on OCBE. ECSR and GWE exert single and serial multiple mediation effects in the link between GSV and OCBE. This study solidifies the conception of GSV as a point from which to measure ECSR, GWE, and OCBE. By exploring ECSR as a serial antecedent to GWE, this study employs a new theoretical angle to measure the single and serial multiple mediation impacts of ECSR and GWE on GSV and OCBE and provides a novel theoretical lens for future research.

1. Introduction

The rapid growth of global tourism has significantly increased the environmental impact of the hospitality industry, which is responsible for approximately 21 % of global CO₂ emissions, extensive consumption of water and energy resources, and substantial production of both recyclable and solid waste (Majeed and Kim, 2023). In response to the pressing concerns from hotel customers and society to adopt eco-friendly practices at hotels, the industry has focused on improving sustainable operations by raising managerial awareness and adopting eco-friendly practices (Elshaer et al., 2022). An organization's environmental performance hinges on how employees engage with and implement organizational-level green work initiatives while performing their individual-level roles at work (Pham et al., 2019; Zhao and Zhou, 2021). This paper investigates how hotels can adopt eco-friendly practices and encourage employees to engage in voluntary, unrewarded extra-green efforts to boost hotel green performance. More precisely,

this paper examines how a hotel's green shared vision (GSV) as an organizational factor can trigger environmental corporate social responsibility (ECSR), which is another organizational factor, to inspire employees' motivation for green work engagement (GWE) and voluntarily contribution to extra-green efforts, i.e., organizational citizenship behavior for the environment (OCBE) as individual factors. Examining organizational-level and individual-level perspectives is important because organizational behavior (OB) perspective of an individual's green behavior may not contribute theoretically and practically at the organizational-level (Boiral et al., 2015). Therefore, this paper connects and examines some organizational-level (GSV and ECSR) and individual-level (GWE and OCBE) perspectives to solidify the holistic understanding of OB.

OCBE shows employees' voluntary extra-green efforts at work that support organizational green initiatives without direct rewards (Boiral, 2009; Kim et al., 2020; Priyankara et al., 2018; Zhao and Zhou, 2021). OCBE includes actions such as conserving resources and waste

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reduction, which align with organizational eco-friendly directions and practices (Zhao and Zhou, 2021). This differs from ECSR, which is a subset of corporate social responsibility (CSR), because ECSR is a corporate-level extra-green efforts in compliance with ethical business standards to push employees to comply with environmental ethics and demonstrate green behavior at work through conserving resources to protect society and environment (Rashid et al., 2014; Yin et al., 2021). Hotel employees play an important role in ensuring the viability of green hotel initiatives for competitive advantage, their engagement in OCBE is key to sustainable hospitality operations.

Leadership's GSV, i.e., a set of eco-friendly initiatives and goals to realize organizational sustainable development (Chang, 2020; Chen et al., 2015), can steer OCBE to boost organizational environmental performance (Elshaer et al., 2022; Zhou et al., 2018). Shared vision is an important motivator in employees' voluntary efforts to achieve organizational success (Boyatzis et al., 2015). Leaders' GSV and commitment to environmental protection act as means through which to develop green objectives and policies and transmit them to hotel employees, which may motivate hotel employees and fuel their GWE. The transmission of GSV to employees fosters skill development and a sense of ethical obligation, and fuels their intrinsic motivation to contribute to environmental goals beyond basic job requirements (Afsar et al., 2019; Chang, 2020; Karatepe et al., 2022; Haldorai et al., 2022). This interconnected relationship highlights the alignment between GSV, GWE, and OCBE.

The ethical undertone of GSV conveys psychological signals that shape employees' environmental attitudes and stimulate engagement in green behaviors, which aligns with signaling theory (Spence, 1973; Yang et al., 2023). ECSR, as a corporate commitment to environmental ethics, signals green objectives to employees, promoting GWE and OCBE (Puppim de Oliveira and Jabbour, 2017; Zhao and Zhou, 2021). GSV's employee-focused nature positively transmits psychological signals to employees and shapes their ethical work attitude alongside motivating them to ethically respond to GSV by participating in ECSR practices (Zhao and Zhou, 2021). Organizational leaders signal their green visionary directions to organization-wide ECSR managers, inducing them to follow and align their strategies with organizational-level green visionary directions for sustainable business in the long run (Zhao and Zhou, 2021). ECSR management shows corporate commitment to environmental ethics, and signals green objectives to employees to promote GWE and OCBE (Puppim de Oliveira and Jabbour, 2017; Zhao and Zhou, 2021). It develops an organizational climate that supports voluntary green efforts among employees (Latif et al., 2022; Zhang et al., 2021). The synergy between ECSR and GSV suggests a pathway through which GSV can drive GWE and OCBE via corporate ethical standards, yet this mechanism still needs to be explored.

Previous studies on OCBE explored coworker advocacy (Kim et al., 2017), employee commitment to realize organizational goals (Paille and Mejia-Morelos, 2014), responsible leadership impact (Afsar et al., 2019), and organizational initiatives in supporting the natural environment (Paille and Raineri, 2015). Although GSV is shown to impact employee commitment and green work behaviors, research has largely overlooked how it serves as a predictor of OCBE. This gap is evident, as existing studies primarily focus on GSV's influence on immediate green behavior metrics without examining OCBE as an outcome, suggesting the need for a more explicit investigation of GSV's long-term influence on voluntary green efforts by employees (Chang, 2020; Yang et al., 2023). Addressing this gap would enhance our understanding of how GSV influences employees' intrinsic motivation to adopt sustainable practices beyond their required roles. Alt and Spitzack (2016) identified shared vision as a moderator between employee involvement, for example, GWE in this paper, and OCBE. However, previous studies failed to address that OCBE is not only a criterion or mediator but also there are antecedents to OCBE that act as mediators between GSV and OCBE, which need to be explored. Although there are traces of evidence depicting the impact of GSV on OCBE (see Chang, 2020; Chang et al., 2019), the literature

mentioned previously has mixed and incoherent findings in terms of the predictor and criterion roles of GSV and OCBE respectively. For example, Aboramadan et al. (2022) focused on leadership style and organizational support, rather than exploring the influence of a shared vision on voluntary green behaviors (OCBE), and do not directly investigate the interplay between GSV and OCBE or consider other potential predictors of OCBE. Similarly, Chang (2020) emphasizes the direct effects of GSV on employee behavior but does not examine indirect pathways or mediating mechanisms. While Chang et al. (2019) link GSV to OCBE, their study is situated in the context of product development, diverging from a hospitality-focused, employee-centric lens. These gaps highlight the need for further research to comprehensively examine the relationship between GSV and OCBE, incorporating multiple mediators and diverse theoretical frameworks to build a more conclusive understanding.

ECSR has been linked with green performance management (Kim, 2022) and employee pro-environment behavior (Zhang et al., 2021), limited studies have examined its role as a mediator in the relationship between GSV and OCBE. In one study, the moderation role of ECSR was examined in the association between the perceived psychological benefits of green hotels and hotel customers' pro-environment behavior (Majeed et al., 2023). Although some studies have created a link between CSR and OCBE (see Luu, 2017), it remains underexplored exactly how ECSR shapes employee voluntary engagement in green activities to support organizational environmental efforts. Specifically, the way ECSR acts as an intermediary in translating GSV into increased OCBE remains underexplored. This gap points to a need for further investigation of how corporate-level green initiatives may influence individual-level green behaviors through ECSR.

Scholars assert that there is a need for more studies of the determinants and outcomes of GWE (Karatepe et al., 2022). Previous studies have often examined ECSR and GWE as mediators in isolation, with GWE viewed as an outcome of green practices rather than as a conduit linking GSV to OCBE (Aboramadan, 2022; Karatepe et al., 2022). For example, previous studies examined the mediation role of GWE to explain the association between different organizational-level predictors and individual-level criterion variables, such as GHRM and employee green behavior (Aboramadan, 2022); management commitment to ecological protection and employee proactive pro-environment behavior (Karatepe et al., 2022); and GHRM and green competitive advantage (Kim et al., 2023a). However, research integrating the individual-level mediation role of GWE and ECSR in the association between GSV and OCBE is notably lacking. Addressing this gap could provide a more nuanced understanding of how GSV impacts voluntary environmental behaviors at individual and organizational levels. However, studies that integrated both ECSR and GWE as sequential mediators between GSV and OCBE remained succinct. A sequential mediation model could illuminate how GSV influences ECSR, which in turn fosters GWE, eventually enhancing OCBE. Understanding the combined influence of ECSR and GWE on OCBE could reveal how organizational and individual-level green initiatives interact to boost environmental performance (Kim et al., 2023a; Latif et al., 2022). While the ability, motivation, and opportunity (AMO) theory (Appelbaum et al., 2000) has been applied to green practices (Kim et al., 2023a; Ari et al., 2020), and signaling theory (Spence, 1973) has been used to explain managerial impact on employee behavior (Yang et al., 2023), the intertwined application of both theories in the GSV-ECSR-OCBE framework has not been fully addressed (for example, see Aboramadan et al., 2022; Chang, 2020; Chang et al., 2019). This gap suggests a need to explore how GSV can signal corporate environmental commitment, triggering ECSR efforts that, in turn, enhance employee OCBE (Puppim de Oliveira and Jabbour, 2017; Zhao and Zhou, 2021).

Based on the AMO theory (Appelbaum et al., 2000) application in different green service business contexts (Ari et al., 2020; Kim, 2022; Kim et al., 2023a), we argue in this paper that GSV can be considered as the launching platform to signal ECSR managers, a phenomenon

supported by the signaling theory (Spence, 1973), to make employees feel that their abilities are being improved, fueling employee motivation for GWE in parallel to voluntarily putting extra-green efforts to boost organizational performance. This shows direct and mediating relationships between GSV, ECSR, GWE, and OCBE, reflecting the intertwined association between AMO (Appelbaum et al., 2000) and signaling (Spence, 1973) theories, which demands research attention. Drawing on the empirical investigation under the spotlights of AMO (Appelbaum et al., 2000) and signaling (Spence, 1973) theories, the purpose of this study is threefold: First, it aims to clarify and examine the direct impact of GSV on OCBE. Second, it seeks to investigate the individual mediating impacts of ECSR and GWE on the relationship between GSV and OCBE. Third, it aims to explore the sequential relationship between ECSR and GWE in the relationship between GSV and OCBE.

Since top leadership role is important in green management (Kim, 2022), we developed a research model (Fig. 1) to investigate how organizational-level (GSV and ECSR) and individual-level (GWE) initiatives may impact OCBE. We expand the conceptual understanding of organizational-level predictors of GWE and OCBE, i.e., GSV and ECSR, to explain the complex woven web of the relationships between GSV and OCBE under the lenses of the AMO (Appelbaum et al., 2000) and signaling (Spence, 1973) theories, which are yet to be explored and understood. This study is the first to bridge the above-mentioned literature gaps. As sustainability becomes a key concern for consumers and stakeholders alike, understanding how hotel employees can be motivated to engage in eco-friendly practices is essential for achieving long-term environmental goals. Exploring GSV and its influence on employee behavior offers valuable insights into enhancing organizational commitment to sustainability. Addressing these issues supports corporate responsibility and nurtures a culture of environmental stewardship within the hospitality sector, making it impactful for future practices and policies.

2. Research model and hypotheses

2.1. Research model

Previous research suggests to incorporate various theories to explain OB issues. It is argued that a single theory may not explain

organizational challenges, and several theories maybe integrated into one study to appropriately streamline, explain, and understand organizational environment and business function (Dönmez et al., 2024). To explore the relationships between GSV, ECSR, GWE, and OCBE among hotel employees, we based our research model (Fig. 1) on AMO (Appelbaum et al., 2000) and signaling (Spence, 1973) theories, incorporating insights from the relevant literature (e.g., Aboramadan, 2022; Chang, 2020; Chang et al., 2020; Kim et al., 2023a). AMO theory (Appelbaum et al., 2000) was used to establish the theoretical connections between GSV and OCBE and between GSV, GWE, and OCBE. Signaling theory (Spence, 1973) was applied to hypothesize the positive relationships between GSV, ECSR, and OCBE. We simultaneously tested GSV's positive direct impact on OCBE and the positive single and serial multiple mediation impacts of ECSR and GWE.

2.2. Hypotheses

2.2.1. GSV and OCBE

GSV reflects management's green strategic directions and commitment to environmental protection to achieve organizational sustainability goals (Chen et al., 2015; Fang et al., 2021; Latif et al., 2022). GSV is an instrumental condition to enable an organization to meet its environmental challenges and manage eco-friendly business operations (Afsar et al., 2019). GSV is a manifestation of top management to encourage employee green initiatives (Elshaer et al., 2022; Meng et al., 2022; Yang et al., 2023). It is a corporate-level green business operation direction that reflects shared goals, commitment, and standards of an organization to steer the green behaviors of employees for a sustainable competitive advantage. OCBE refers to voluntary, extra-green efforts, personal green initiatives, and proactive employee eco-friendly behaviors that help an organization realize its green goals (Luu, 2017; Umrani et al., 2020; Wiradirja et al., 2020). It is a discretionary social behavior and mirrors eco-friendly approaches at work, for example, gathering more environmental knowledge, implementing green practices, reducing waste, conserving resources, and depolluting the work environment. Despite improving eco-friendly performance, OCBE helps improve organizational green reputation (Elshaer et al., 2022). Previous studies have articulated a theoretical understanding of extra-role employee green behavior, i.e., pro-environment behaviors beyond an

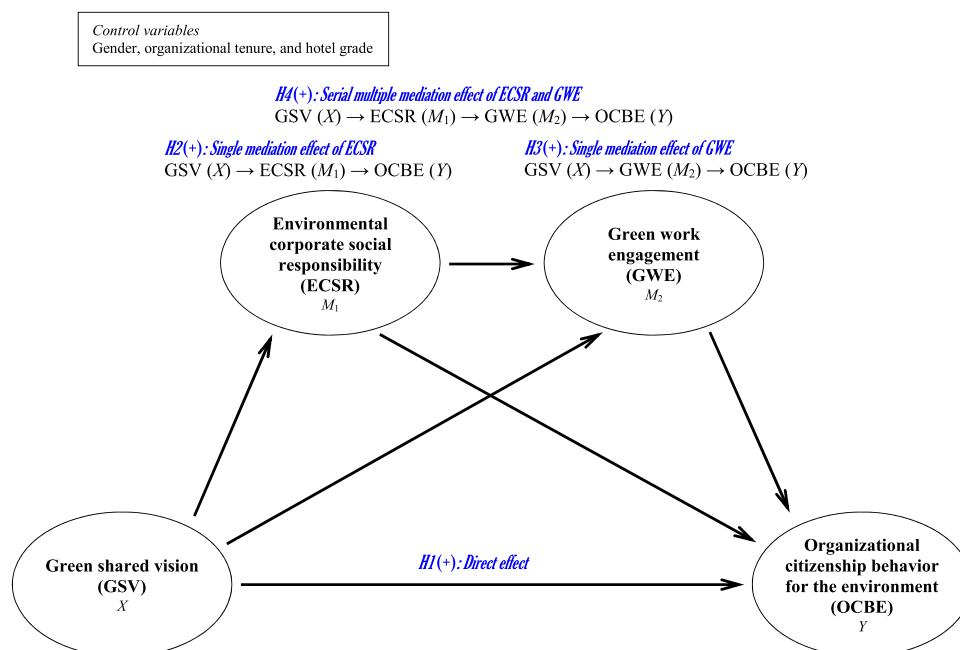


Fig. 1. Research model.

employee's formal duties (Aboramadan, 2022), which is closely linked to OCBE in this research. However, OCBE is often unrecognized and unrewarded by an organization (Kim et al., 2020; Pham et al., 2019; Zhao and Zhou, 2021).

Umrani et al. (2020) believe that employee willingness to perform green actions beyond those dictated by a formal organizational reward system depends on management's green vision and degree of commitment to environmental sustainability. The AMO (Appelbaum et al., 2000) and signaling (Spence, 1973) theories jointly explain how GSV impacts OCBE. AMO theory (Appelbaum et al., 2000) posits that employees are most likely to engage in desired behaviors when they have the ability, motivation, and opportunity to do so (Appelbaum et al., 2000). Under the spotlight of AMO theory (Appelbaum et al., 2000), organization-level green initiatives (GSV) facilitate employee engagement in voluntary pro-environmental activities (OCBE) by enhancing their abilities (through green skill development), motivation (via alignment with environmental values), and opportunities (with organizational support). Within this context, GSV is a motivator and opportunity provider that fosters employees' eco-friendly initiatives, enabling them to align their personal values with organizational sustainability objectives. A strong GSV communicates a clear green strategy, enabling employees to develop eco-friendly skills and knowledge. When management actively supports environmental training and skill development, employees gain the ability to effectively engage in green practices, enhancing OCBE (Meng et al., 2022). Aligning employee skills with environmental goals ensures that they have the knowledge and capabilities to make meaningful environmental contributions at work. GSV significantly motivates employees by establishing environmental sustainability as a core organizational value, instilling a sense of pride and purpose. Management's commitment and directions to improve employee abilities motivate employees to reciprocate it by putting additional efforts to achieve organizational goals and boost sustainable performance (Kim et al., 2023a). This intrinsic motivation stems from employees perceiving GSV as both an organizational value and a pathway to personal and professional growth.

Signaling theory (Spence, 1973) argues that organizations can convey information about their intentions, values, and priorities through observable signals. In the context of GSV, the organization's leadership communicates its commitment to environmental goals and sustainability practices as a signal to employees. This communication acts as a credible assurance of the organization's dedication to fostering a green culture, influencing employees' perceptions and attitudes. Signaling theory (Spence, 1973) complements the premises of the AMO theory (Appelbaum et al., 2000) in support of the impact of GSV on OCBE by illustrating how GSV acts as a signal of management's commitment to environmental goals, developing trust (an implied notion) among employees. This trust motivates employees to reciprocate by voluntarily contributing to sustainability initiatives, aligning their efforts with organizational objectives. The psychological and ethical signals inherent in GSV encourage employees to align their individual green behaviors with broader organizational goals. GSV serves as a strong and consistent signal because it is embedded in organizational strategy and visibly supported by leadership actions. For example, when leaders articulate a GSV through policies, goals, and environmental practices, it reassures employees that sustainability is not merely a peripheral concern but a core organizational value. This consistency in signaling builds trust, aligning employees' values with the organization's eco-friendly objectives and fostering a sense of shared purpose.

The ethical undertones of GSV as a signal further enhance its impact. Employees perceive the green vision as an organizational commitment to broader individual (i.e., ability), societal, and environmental well-being, which resonates with their intrinsic motivation to engage in green practices. This moral alignment compels employees to reciprocate the organization's efforts by adopting OCBE, creating a virtuous cycle of sustainability-focused behavior. Employees are more likely to engage in OCBE when they perceive a robust green signal (i.e., commitment to

protect the environment) from top management, viewing their green contributions as meaningful and valued by the organization (Meng et al., 2022; Yang et al., 2023; Zhao and Zhou, 2021). This intrinsic motivation to support environmental goals further amplifies their dedication to voluntary green efforts. Employees' motivation to voluntarily take on additional work is susceptible to top management's visionary support and directions (i.e., green signals) to elevate employee knowledge and working abilities, which are considered as an opportunity by employees to become more competitive in the market (Kim, 2022; Meng et al., 2022).

GSV signals the necessary opportunities for employees to contribute to environmental objectives by forging a supportive environment for green initiatives. When management signals a clear GSV, it creates an atmosphere where employees feel empowered to make eco-friendly choices, regardless of whether such behaviors are formally recognized or rewarded (Kim, 2022). Leaders' green visionary signals allow employees to demonstrate their environmental commitment, helping build a sustainable competitive advantage (Latif et al., 2022; Zhao and Zhou, 2021). Employees interpret top management's signals as a call to action, understanding that their contributions to environmental sustainability will be valued and appreciated. This motivates them to engage in voluntary pro-environmental behaviors—such as conserving resources or participating in green initiatives—that support organizational sustainability goals, even without explicit rewards or recognition. Leaders' communicated green visionary directions may inspire employees to develop green working abilities, and this opportunity triggers employees' intrinsic motivation to foster strong bond with organization and support organizational sustainable initiatives through voluntary green efforts (Yang et al., 2023). These premises align with the AMO (Appelbaum et al., 2000) and signaling (Spence, 1973) theories because top leaders' visionary practices are equivalent to green signals that aim to influence, motivate, and boost employee behavior at work in realizing organizational green goals for a competitive advantage. Through the lens of AMO (Appelbaum et al., 2000) and signaling (Spence, 1973) theories, this relationship highlights the critical role of management in promoting eco-friendly employee behaviors that contribute to long-term sustainability.

Given recent global environmental protection efforts, employees expect organizations to have a GSV, which encourages them to demonstrate green behavior at work (Latif et al., 2022) and motivates them to perform beyond their pre-defined role, i.e., OCBE, to support organizational green efforts (Chang, 2020). Leaders' failure to demonstrate GSV may dampen employee green inspiration and decrease voluntary green efforts (Afsar et al., 2019). Scholars have attempted to discover the positive impact of GSV on OCBE in multiple contexts, such as Taiwanese manufacturing (Chang et al., 2019) and textile industries (Wiraditja et al., 2020). Priyankara et al. (2018) established the positive effect of a leader's support for the environment, which arguably resembles GSV as defined in this paper, on OCBE in the context of the textile and apparel manufacturing industry. Understanding how a hotel's GSV can drive OCBE is important for improving both operational efficiency and corporate reputation. As consumers become more eco-conscious, developing a culture of eco-friendly practices among employees aligns with societal expectations and provides hotels with a competitive edge. By exploring the relationship between GSV and OCBE, this paper attempts to contribute to broader discussions about organizational commitment to sustainability and the instrumental role of leadership in motivating green initiatives within the hospitality sector. In line with the preceding arguments and theoretical underpinnings, we hypothesize that:

H1. GSV has a positive effect on OCBE.

2.2.2. ECSR as a single mediator

CSR in hospitality includes broader notions of social and environmental initiatives in terms of community engagement, ethical labor practices, and basic environmental measures like waste reduction,

endured by compliance and reputation enhancement (Kang and Lee, 2022). In contrast, ECSR, which is a subset of CSR, refers to proactive and explicit environmental actions that extend beyond regulatory compliance to develop deep organizational commitment to environmental protection, often with more intensive sustainability metrics and reporting (Wang and Kim, 2021). CSR initiatives focus on general social responsibility and typically lack focused environmental strategies and employee engagement. However, ECSR involves more measurable environmental initiatives that signal to engage organizational management and employees as active participants in realizing organizational green performance (Wang and Kim, 2021; Xu and Huang, 2022).

From an AMO theory (Appelbaum et al., 2000) perspective, GSV as leadership's commitment to support ECSR managers in enhancing employees' abilities and motivation embed sustainability into corporate practices, providing a framework for employees to contribute. These initiatives create opportunities for employees to engage in OCBE by establishing an organizational culture that values environmental responsibility. This cultural shift aligns organizational and employee goals, developing green behaviors that exceed formal job requirements. ECSR acts as a bridge between GSV and OCBE by operationalizing an organization's environmental priorities into structured and actionable corporate practices. From the AMO theory (Appelbaum et al., 2000) perspective, ECSR management improves employees' capabilities to engage in environmental initiatives through resources, training, and knowledge-sharing opportunities. By integrating sustainability into the organizational framework, ECSR managers provides employees with the tools and motivation to voluntarily exceed their formal job roles and actively contribute to green practices.

Signaling theory (Spence, 1973) adds depth to AMO theory (Appelbaum et al., 2000) by portraying ECSR as an organizational signal of authenticity and commitment to sustainability. This signal enhances employee trust and alignment with the organization's green vision, motivating them to engage in voluntary green actions. Using signaling theory (Spence, 1973), scholars have established that hotel leadership's GSV and commitment to environmental protection are signals to hotel managers of corporate initiatives, i.e., ECSR managers, protecting the natural environment and society, and that ECSR managers transmit those signals to hotel employees to fuel green efforts beyond those required as part of their job duties (Karatepe et al., 2022; Latif et al., 2022), which are comparable to OCBE in this study. Organizational GSV transmits signals of the environmental protection approach of the top management to organizational ECSR managers (Chang et al., 2020), who articulate their vision and motivate employees to perform extra green activities, i.e., OCBE, to realize organizational green goals (Puppim de Oliveira and Jabbour, 2017). This study adopts signaling theory (Spence, 1973) to explore how GSV signals ECSR managers and employees, showing organizational commitment to environmental stewardship and encouraging OCBE (Chen et al., 2021). We highlight how ECSR when pushed by GSV, signals strong organizational values that encourage employees to engage in OCBE.

Given the global environmental degradation and customer pressure on hotels to follow green business practices (Kim et al., 2023a; Majeed and Kim, 2023), hotel employees are also responsive to hotel ECSR and so may put extra effort into supporting hotel green efforts (Karatepe et al., 2022). GSV acts as a signal of the organization's environmental dedication to ECSR managers, who pass on these signals to employees to develop trust and transparency between organizational leaders and employees (Kim et al., 2023b), which strengthens employees' voluntary participation in environmental activities, leading to OCBE (Yun and Park, 2022). When leaders of an organization consistently communicate GSV to employees through ECSR management, it makes employees feel that the organization is dedicated to fulfilling the prophecy of ECSR (Zhou and Wang, 2024). This signaling effect motivates employees to contribute to environmental sustainability through OCBE, as they recognize that their actions align with the organization's long-term objectives. Chang et al. (2020) reveal the mediation role of ECSR

between GSV and employee environmental performance. Scholars argue that ECSR managers, who are signalers of environmental protection visionary plans, receive green signals from organizational top management and act as a bridge between them and employees, transmitting their green vision to boost employee environmental performance (Chang et al., 2020). In this study, employee environmental performance is comparable to OCBE because both concepts indicate employee performance in realizing organizational green goals. In hospitality and tourism, Zhang et al. (2021) acknowledged the mediation role of ECSR perception between environmental knowledge and employee pro-environment behavior. GSV articulates organizational top leadership's green initiatives (Bhutto et al., 2021; Latif et al., 2022), and employee green performance is a conduit towards employee pro-environmental behavior (Kim et al., 2023a). Drawing on the works of Zhang et al. (2021) and the arguments of Bhutto et al. (2021) and Latif et al. (2022), we deduce that ECSR perception is comparable to ECSR and that ECSR plays a mediation role between GSV and OCBE as presented in this study. By examining how GSV signals organizational commitment to environmental stewardship and influences ECSR initiatives, we can better understand how to trigger employees' eco-friendly behaviors beyond their formal duties. It is important to understand these relationships to reveal practical insights for hospitality leaders seeking to enhance their sustainability efforts and improve organizational performance through active employee participation in environmental initiatives. Drawing on the above, we propose the following hypothesis:

H2. ECSR mediates the positive relationship between GSV and OCBE, such that GSV positively influences ECSR, which subsequently leads to a positive effect on OCBE.

2.2.3. GWE as a single mediator

Employee GWE reflects employee's cognitive-affective state, dedication, and immersion in green work-related tasks (Bhutto et al., 2021; Çop et al., 2020; Kim et al., 2023a). In sustainable tourism, scholars assert that employee motivation and work engagement are influenced by the guidance of organizational leadership and their commitment to organizational goals (Bhutto et al., 2021). GSV positively influences GWE by developing eco-friendly culture that aligns organizational goals with individual values (Lee et al., 2023). The AMO theory posits that employees are more likely to engage in green work behaviors when they feel motivated, capable, and supported in their environmental efforts (Ari et al., 2020; Kim et al., 2022). When an organization emphasizes GSV, it provides a collective sense of purpose, motivating employees to participate actively in eco-friendly practices that contribute to sustainability goals (Su et al., 2023). For example, green training and resources offered within a GSV framework develop employees' green skills (ability) and enhance their motivation to adopt sustainable practices, ultimately leading to higher GWE (Lee and Sung, 2022). Once employees are engaged in green work, GSV further influences OCBE by encouraging them to take on additional eco-friendly responsibilities beyond formal job duties. In this context, GWE serves as a motivational mediator that translates GSV into action, supporting extra-role behaviors and contributing to the environmental performance of organization (Afsar and Rehman, 2023).

Signaling theory (Spence, 1973) also supports the theoretical premises of AMO theory (Appelbaum et al., 2000) in providing a compelling framework to understand how GSV influences GWE, which in turn facilitates OCBE. Signaling theory (Spence, 1973) suggests that organizations communicate their intentions, priorities, and values through observable signals (Spence, 1973). GSV, as a visible manifestation of an organization's commitment to sustainability, acts as a strong signal to employees about the importance of environmental stewardship within the organization. This signal creates clarity and alignment between organizational goals and employees' roles, motivating them to engage deeply with environmental initiatives. When leaders effectively communicate a GSV through policies, goals, and actions, it signals to

employees that sustainability is a core organizational priority (Latif et al., 2022). Employees interpret this signal as an opportunity to align their efforts with these goals, leading to heightened GWE. The organizational emphasis on environmental values fosters a sense of shared purpose, inspiring employees to become more engaged, enthusiastic, and immersed in green tasks (Kim et al., 2023a; Bhutto et al., 2021).

GWE reflects employees' cognitive and emotional investment in sustainability initiatives, fueled by their belief in the organization's commitment to long-term environmental goals. GSV also conveys psychological and ethical signals that enhance trust and transparency within the organization. Employees perceive GSV as evidence of the organization's integrity and alignment with societal expectations for environmental responsibility. This perception strengthens their intrinsic motivation to reciprocate these values by engaging in green tasks beyond their formal job requirements (Su et al., 2017). Such heightened engagement is crucial because it transforms GWE into a mediator that connects the organization's environmental vision (GSV) with voluntary green behaviors (OCBE).

Employees committed to GSV principles may go beyond task-specific duties to reduce waste, conserve resources, or help colleagues engage in sustainable practices (Su et al., 2023). This positive effect of GSV on OCBE, via GWE, emphasizes the role of a shared environmental vision in motivating employees to voluntarily support organizational sustainability. Kim et al. (2023a) adopted AMO theory (Appelbaum et al., 2000) to delineate the mediation role of GWE, and found that hotel green knowledge sharing reflects hotel management's ability and commitment to meet environmental protection goals, which motivates hotel employees to demonstrate GWE behavior at work and perform extra green behaviors to meet hotel objectives, i.e., hotel employee OCBE. Hotel green knowledge sharing is comparable to GSV because GSV also reflects organizational leadership's direction and commitment to addressing environmental challenges. Understanding how GSV shapes GWE, which in turn determines OCBE, highlights the role of leadership in establishing a sustainable workplace culture. This research is important because it contributes to environmental sustainability goals and enhances employee motivation and organizational performance, aligning individual values with broader corporate responsibility initiatives. Given all of this, we hypothesize that:

H3. GWE mediates the positive relationship between GSV and OCBE, such that GSV positively influences GWE, which subsequently leads to a positive effect on OCBE.

2.2.4. ECSR and GWE as serial multiple mediators

Previous studies support the assertion that GSV increases employees' sense of environmental responsibility, leading to green behaviors, including those outside their job duties (Fang et al., 2021). Considering the findings of Fang et al. (2021), AMO (Appelbaum et al., 2000) and signaling (Spence, 1973) theories, and the previous discussion of the interlinked roles of GSV and ECSR (H2) in connection to GSV and GWE (H3) in determining employee OCBE, we deduce that the green visionary policies of top management transmit ability-enhancing and competitiveness-improving motivational signals to employees (Ari et al., 2020; Chang et al., 2020; Karatepe et al., 2022; Kim et al., 2023a) such as ECSR managers in this study.

Scholars note that hotel employees who receive these signals from hotel management (for example, ECSR managers in this study) transmit them to fellow employees, making them motivated and highly engaged in pro-environment activities (Karatepe et al., 2022; Latif et al., 2022). Employee perception of CSR impacts their engagement and performance in regard to green behaviors (Shah et al., 2020). Addressing the impacts of GWE on employee attitude and behavior, some studies note that GWE is a motivational mechanism stimulated by employees' positive exchanges with employers, which increases employees' voluntary green activities (Aboramadan, 2022), i.e., OCBE, responding positively to a hotel's sustainability objectives. These arguments help to articulate the intervening serial mechanisms of ECSR and GWE in the impact of GSV

on OCBE.

ECSR and GWE often support GSV's influence on OCBE as sequential mediators. For example, scholars adopted AMO theory (Appelbaum et al., 2000) to explore the serial multiple mediation impacts of "green knowledge sharing and GWE" in the association between "GHRM and green competitive advantage" (Kim et al., 2023a). Green knowledge sharing is comparable to ECSR in this study, as ECSR managers share green corporate socially responsible knowledge and organizational expectations of environmental commitment with fellow employees to stimulate GWE (Karatepe et al., 2022). GHRM is comparable to GSV in this study, as GHRM and GSV are organizational-level communications that transmit green policies and strategies to employees. Employee green performance is an implied mechanism that influences hotels' green competitive advantage (Kim et al., 2023a), comparable to OCBE in this study. ECSR reflects the organization's overarching commitment to environmental sustainability, providing employees with a supportive framework for their green efforts. GSV revitalizes this commitment by establishing an internal culture where sustainability is a shared priority, which, in turn, leads to GWE through ECSR. GWE then becomes the final pathway through which individual employees engage in OCBE, showing how GSV motivates and enables environmentally supportive behaviors. This sequential mediation aligns with AMO theory (Appelbaum et al., 2000) by ensuring that employees have the necessary ability, motivation, and opportunity to engage in OCBE (Kim et al., 2022). Based on the above arguments, work of Kim et al. (2023a), and extending from the extant theoretical underpinnings of GSV, ECSR, GWE, and OCBE (Aboramadan, 2022; Fang et al., 2021; Karatepe et al., 2022; Latif et al., 2022), we postulate the serial multiple mediation roles of ECSR and GWE in the link between GSV and OCBE in our work. Thus, we hypothesize the following:

H4. ECSR and GWE sequentially mediate the positive relationship between GSV and OCBE, such that GSV positively influences ECSR, which subsequently leads to a positive effect on GWE, ultimately resulting in a positive impact on OCBE.

3. Methods

3.1. Participants and data garnering

We garnered responses from hotel staff members working at upper-upscale and luxury hotels in Seoul, South Korea. As of the data collection date in May 2022, 41 upper-upscale and 30 luxury hotels were in business in the Seoul Metropolitan Area (see www.hotelrating.or.kr). Among them, the heads of human resources (HR) for 15 hotels agreed to participate in the survey. We delivered a survey packet to each head of HR and provided them with an envelope in which to return the responses, a gift token, and a summary document stating that the questionnaires must be returned directly to the questionnaire administrator in the sealed envelope. The survey was conducted over approximately three weeks, from May 20 to June 9, 2022. The HR directors handed out 450 questionnaires (30 questionnaires per hotel) to employees, and 397 questionnaires were returned.

We excluded 7 surveys with missing values and 32 surveys that contained a problem with the normality of data through the data cleaning and screening processes. Since the |skewness| (OCBE3 −1.555 to GWE3 −0.657) and |kurtosis| values (GSV2 −0.075 to OCBE3 1.450) did not exceed two and seven (West et al., 1995), respectively, there was no problem with univariate normal distribution. We checked the Mahalanobis distance value and deleted 19 cases. The Mardia (1970)'s coefficient (108.207) was $< p(p+2)$ ($p = \#$ of indicators = 17; 323). Hence, the multivariate normal distribution was also satisfied. Therefore, we retained 358 surveys (397 initially, minus 39 excluded) for the final analysis, offering a 79.6 % (358/450) available response rate.

The recommended minimum sample size for structural equation modeling using Free Statistics Calculators 4.0 program was 137 [anticipated effect size (f^2) = 0.30 medium effect size; desired statistical

power (1 − β error probability) level = 0.80; number of latent variables = 4 (GSV, ECSR, GWE, and OCBE); number of observed variables = 20 (4 items for GSV, 4 items for ECSR, 6 items for GWE, 3 items for OCBE, and 3 control variables); and α error probability level = 0.05]. Consequently, the sample size of this work, $N = 358$, was deemed acceptable (Soper, 2024). Respondents' profiles are presented in Table 1.

3.2. Common method bias: procedural treatments

Common method variance (CMV) refers to “variance that is attributable to the measurement method rather than to the construct of interest” (Podsakoff et al., 2003, p. 879). CMV can introduce systematic measurement errors, leading to biased estimates of the “true” relationships between constructs. This bias in estimates due to CMV is known as common method bias (CMB). While CMV might be present, it only becomes problematic when CMB reaches a significant level. Therefore, the issue is not CMV's presence per se but CMB's extent (Bae, 2014). CMB arises from artificial covariance between variables when both predictor and criterion variables are measured using the same method (e.g., questionnaire) and respondent. This bias negatively impacts construct validity and reliability, distorting research results by either inflating or attenuating the “true” relationship between constructs (Baumgartner and Steenkamp, 2001; Doty and Glick, 1998; Park et al., 2007; Podsakoff et al., 2003; Podsakoff et al., 2024; Williams and Anderson, 1994; Williams et al., 2010).

To minimize potential CMB, this empirical study applied several procedural remedies throughout the research design and data collection phases, following recommendations by Podsakoff et al. (2003) and practices established in prior studies (e.g., Kim and Karatepe, 2023; Kim et al., 2023; Olorunsola et al., 2024). First, to reduce the likelihood that respondents would perceive connections between the measured

variables, we psychologically separated them by avoiding any mention of causal relationships on the questionnaire cover page. Second, to encourage honest responses and minimize evaluation apprehension and social desirability bias, the survey was conducted anonymously, with respondents receiving envelopes to seal their completed questionnaires for confidentiality. The cover page also emphasized, “There are no right or wrong answers.” Third, variables were presented in reverse order (OCBE, GWE, ECSR, and GSV) to help prevent respondents from discerning causal relationships (Fig. 1). This order arrangement also aimed to mitigate priming effects by altering the sequence of items related to each variable. Finally, different scale endpoints and formats were used between variables, employing methodological separation to reduce CMB related to consistent scale endpoints and anchoring effects.

3.3. Measurement

The study examined GSV through four items adopted from Chen et al. (2015), e.g., “My hotel has commonalities in environmental goals.” Participants responded to these items on a 7-point Likert scale (“1 = strongly disagree” to “7 = strongly agree”). ECSR was measured with four items adapted from Park and Levy (2014) and Türker (2009), e.g., “My hotel participates in activities which aim to protect and improve natural environment.” A 5-point Likert scale (“1 = strongly disagree” and “5 = strongly agree”) was used. This study measured GWE by adapting six items from Aboramadan (2022), e.g., “My environmental-related tasks inspire me.” A 7-point Likert scale (“0 = never” to “7 = always”) was utilized. A total of ten scale items for OCBE (second-order factor structure; the three first-order factors: eco-initiatives [three items; e.g., “In my work, I weigh the consequences of my actions before doing something that could affect the environment”], eco-civic engagement [four items; e.g., “I actively participate in environmental events organized in and/or by my hotel”], and eco-helping [three items; e.g., “I spontaneously give my time to help my colleagues take the environment into account in everything they do at work”]) were adopted from Boiral and Paille (2012). Participants responded to these items on a 5-point Likert scale (“1 = completely disagree” to “5 = completely agree”). Appendix A shows a complete list of scale items. We included gender, organizational tenure, and hotel grade as control variables to rule out potential confounding influences (Kim et al., 2023a).

4. Findings

4.1. CMB detecting and controlling: statistical treatments

As mentioned previously, this study implemented four procedural remedies recommended by Podsakoff et al. (2003) and widely adopted in prior research (e.g., Kim and Karatepe, 2023; Kim et al., 2023; Olorunsola et al., 2024). However, due to the nature of the survey method and research model, it remains somewhat susceptible to biases. To further address CMB, the authors used an unmeasured common latent factor (UCLF) technique to detect and control for CMB across various statistical dimensions. Because response sources and survey timing were not separated for study variables—an otherwise effective method for managing CMB during data collection (Park et al., 2007; Podsakoff et al., 2003; Podsakoff et al., 2024)—the UCLF approach was utilized. This technique enables researchers to identify and adjust for CMB by capturing variance due to measurement artifacts rather than actual relationships among variables (Podsakoff et al., 2003; Podsakoff et al., 2024; Williams et al., 2003).

The results indicated that, first, the average variance explained by each latent factor in describing observed variables was 70.9 %, whereas the UCLF accounted for only 6.5 % of the observed variance. Given that the typical average variance attributed to other respondents is approximately 7 % (Min et al., 2016), the variance explained by the UCLF is relatively low. Second, a significant difference was observed between the model with UCLF and the model without UCLF ($\Delta\chi^2[df] = 69.715$

Table 1
Subject profile ($N = 358$).

Characteristics	<i>N</i>	%
<i>Gender</i>		
Male	185	51.7
Female	173	48.3
<i>Age</i>		
Younger than 30	102	28.5
30–39	140	39.1
40–49	82	22.9
Over 49	34	9.5
<i>Education</i>		
High school	53	14.8
Two-year college	158	44.1
Four-year college	110	30.7
Graduate degree	37	10.3
<i>Organizational tenure</i>		
Less than 5 years	211	58.9
5–9 years	72	20.1
10–14 years	41	11.5
More than 14 years	34	9.5
<i>Job position</i>		
First-line level	184	51.4
Supervisor level	81	22.6
Assistant manager level	59	16.5
Manager level or above	34	9.5
<i>Department</i>		
Front desk	67	18.7
Housekeeping	62	17.3
Food & Beverage	76	21.2
Cooking	67	18.7
Sales & Marketing	19	5.3
Human resources	17	4.7
Finances	12	3.4
Maintenance	29	8.1
Other	9	2.5
<i>Hotel grade</i>		
Four-star	147	41.1
Five-star	211	58.9

[17], $p < 0.001$), with the model fit indices for the UCLF model aligning more closely with prior research findings (e.g., Kim et al., 2023; Williams and Anderson, 1994). Third, several measurement items showed a difference of greater than |0.20| (Serrano-Archimi et al., 2018) in standardized factor loadings between the model with UCLF and the model without UCLF. Lastly, the positive impact of GSV on GWE was not significant in the structural model without UCLF but became significant in the model with UCLF, suggesting that the empirical relationship between GSV and GWE was negatively affected by CMB (Bae, 2014; Podsakoff et al., 2024; Williams and Anderson, 1994; Williams et al., 2003). This change implies that controlling for CMB reveals a more accurate relationship, as the constructs are cognitively interconnected at the individual level, which increases the degree of covariance between them (Kim, 2012). These findings align with arguments by various scholars (e.g., Baumgartner and Steenkamp, 2001; Doty and Glick, 1998; Park et al., 2007; Podsakoff et al., 2003, 2024; Williams and Anderson, 1994; Williams et al., 2010) that uncontrolled method variance can bias both construct reliability and validity, as well as parameter estimation, distorting the “true” relationship between constructs. Therefore, we controlled (i.e., accounted) for the 6.5 % of method variance by including the UCLF in both the measurement (Table 2) and structural models (Fig. 2) (Bae, 2014; Podsakoff et al., 2024; Williams et al., 2003).

4.2. Measurement model testing

Structural equation modeling was conducted using AMOS 24.0. As in prior works (e.g., Kim and Karatepe, 2023), the domain-representative parceling technique for the second-order factor of OCBE was used for model parsimony and improvement of model fit statistics. Three parcels

Table 2
Reliability and convergent validity testing.

Constructs and indicators	SFLs (t-values)	AVE	CR	Alpha
<i>Green shared vision (GSV)</i>		0.821	0.948	0.955
GSV1	0.912 (32.931)			
GSV2	0.847 (26.918)			
GSV3	0.949			
GSV4	0.912 (34.480)			
<i>Environmental corporate social responsibility (ECSR)</i>		0.672	0.891	0.924
ECSR1	0.768 (13.517)			
ECSR2	0.749 (10.564)			
ECSR4	0.789 (15.559)			
<i>Green work engagement (GWE)</i>		0.649	0.917	0.930
GWE1	0.854			
GWE2	0.732 (16.667)			
GWE3	0.776 (18.399)			
GWE4	0.854 (21.668)			
GWE5	0.778 (18.327)			
GWE6	0.832 (20.012)			
<i>Organizational citizenship behavior for the environment (OCBE)</i>		0.731	0.890	0.919
OCBE1	0.774 (21.285)			
OCBE2	0.903			
OCBE3	0.881 (23.282)			

Note: CR = construct reliability; AVE = average variance extracted; SFLs = standardized factor loadings.

as three indicators were generated for the three first-order factors of OCBE: eco-initiatives, eco-civic engagement, and eco-helping. Thus, the measurement model consists of four factors and seventeen indicators (Table 2). The four-factor model (Table 3) is the best suited to the data with $\chi^2_{df} = 207.187_{[96]}$; normed $\chi^2 = 2.158$; root mean square error of approximation (RMSEA) [90 % CI] = 0.057 [0.046; 0.068]; normed fit index (NFI) = 0.964; comparative fit index (CFI) = 0.980; and Akaike information criterion (AIC) = 355.187 in comparison to other models (Hair et al., 2010). In short, the distinctiveness of four constructs under examination was confirmed.

As shown in Table 2, all standardized factor loadings at $p < 0.001$ were significant. By following Fornell and Larcker's (1981) criteria, we determined convergent validity [i.e., average variance extracted (AVE) scores > 0.50]. The Heterotrait-Monotrait ratio of correlations (Henseler et al., 2015) values (Table 4) did not exceed 0.850 (Clark and Watson, 1995); thus, strict discriminant validity was satisfied.

As the results (Table 2) show, all composite reliability scores were > 0.70 (Fornell and Larcker, 1981). Additionally, all coefficient alphas > 0.70 and thus confirmed the reliability (Hair et al., 2010). Table 5 presents mean, standardized deviation, and correlations of observed variables.

4.3. Structural model testing

As described in the section on CMB detection and control through statistical treatments, this study determined that CMB was present and conducted analyses while controlling for it within the structural model (Bae, 2014; Podsakoff et al., 2024; Williams et al., 2003). A significant difference emerged when comparing the structural model with CMB control to the one without it. In the uncontrolled model, the path from GSV to GWE was insignificant, but in the model that controlled for CMB, this path became significant. Thus, the change in the significance of this structural parameter indicates that CMB should be considered a factor in the data (Bae, 2014; Podsakoff et al., 2024). Consequently, the results from the model that controlled for CMB are likely more accurate than those from the analysis that did not control for it (Bae, 2014; Podsakoff et al., 2024; Williams et al., 2003).

This study used 5000 bootstrap resamples at 95 % of bias-corrected confidence interval (BCCI) for the user-defined estimands function to verify the structural model containing single and serial multiple mediation effects (Preacher and Hayes, 2008). The structural model (Fig. 2) showed acceptable or excellent model fit statistics: $\chi^2_{df} = 300.003_{[138]}$; normed $\chi^2 = 2.174$; RMSEA [90 % CI] = 0.057 [0.048; 0.066]; NFI = 0.949; and CFI = 0.971 (Hair et al., 2010). All hypotheses were supported. The results accounted for 3.3 %, 22.3 %, 16.7 %, and 41.6 % of the variances in GSV, ECSR, GWE, and OCBE, respectively.

4.3.1. Direct effect

The results in Fig. 2 revealed that GSV had a significant positive effect on OCBE ($\beta = 0.171$, $t = 3.499$, $p < 0.001$). Hence, H1 is supported.

4.3.2. Single mediation effects

In the link between GSV and OCBE, the 95 % BCCI around the indirect impact via ECSR ($b = 0.061$, 95 % BCCI = [0.037; 0.108]; $BSE = 0.019$) (a_1b_1) or GWE ($b = 0.009$, 95 % BCCI = [0.002; 0.136]; $BSE = 0.028$) (a_2b_2) did not contain zero and was significant and positive. Thus, H2 and H3 are supported. Putting the above results together, between GSV and OCBE, as both mediation (a_1b_1 , a_2b_2) and direct effects were found and the direction was positive, ECSR and GWE, respectively, played complementary mediation roles (Zhao et al., 2010).

4.3.3. Serial multiple mediation effect

The Serial multiple mediation effect ($b = 0.007$, 95 % BCCI = [0.001; 0.025]; $BSE = 0.008$) of ECSR and GWE in the relationship between GSV and OCBE ($a_1d_{21}b_2$) was significant and positive because the 95 % BCCI did not contain zero. Hence, H4 is supported. Between GSV and OCBE, as

Control variables	γ
Hotel grade $\rightarrow$ GSV	0.171**
Hotel grade $\rightarrow$ ECSR	0.209***
Organizational tenure $\rightarrow$ ECSR	0.212***
Organizational tenure $\rightarrow$ OCBE	0.242***

H4(+): Serial multiple mediation effect of ECSR and GWE

GSV (X) $\rightarrow$ ECSR (M_1) $\rightarrow$ GWE (M_2) $\rightarrow$ OCBE (Y) $a_1 d_{21} b_2$
 $b = 0.007$; 95% $BCCI = [0.001; 0.025]$; $BSE = 0.008$

H2(+): Single mediation effect of ECSR

GSV (X) $\rightarrow$ ECSR (M_1) $\rightarrow$ OCBE (Y) $a_1 b_1$
 $b = 0.061$; 95% $BCCI = [0.037; 0.108]$; $BSE = 0.019$

H3(+): Single mediation effect of GWE

GSV (X) $\rightarrow$ GWE (M_2) $\rightarrow$ OCBE (Y) $a_2 b_2$
 $b = 0.009$; 95% $BCCI = [0.002; 0.136]$; $BSE = 0.028$

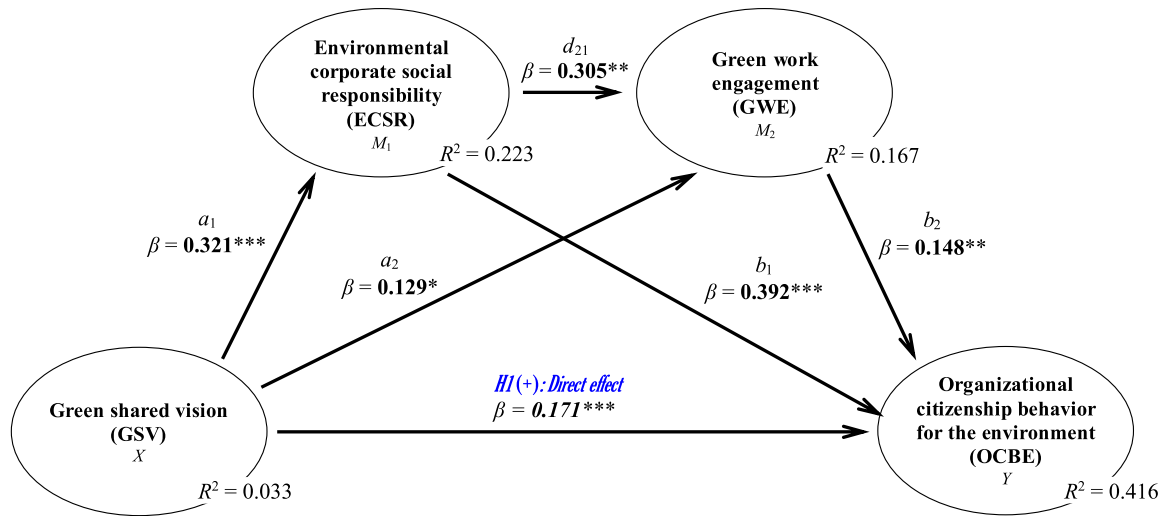


Fig. 2. Structural model testing. Notes: UCLF was added to control for CMV, and indicators were allowed to load on their theoretical constructs and on a UCLF. $BCCI$ = bias-corrected confidence interval; b = unstandardized coefficient; β = standardized coefficient; BSE = bootstrap standard error; UCLF = unmeasured common latent factor; CMV = common method variance. t -values (one-tailed test): $t > 1.65$, $*p < 0.05$, $t > 2.33$, $**p < 0.01$, and $t > 3.09$, $***p < 0.001$.

Table 3
Measurement model comparison.

Models	$\chi^2_{(df)}$	Normed χ^2	RMSEA [90 % CI]	NFI	CFI	AIC	Model comparison; $\Delta\chi^2_{(df)}$
1	207.187 _[96]	2.158	0.057 [0.046; 0.068]	0.964	0.980	355.187	
2	258.569 _[99]	2.612	0.067 [0.057; 0.074]	0.955	0.971	400.569	1 and 2; 51.382 _[3] ***
3	807.649 _[101]	7.997	0.140 [0.131; 0.149]	0.858	0.873	945.649	1 and 3; 600.462 _[5] ***
4	2187.704 _[103]	21.240	0.238 [0.229; 0.247]	0.615	0.624	2321.704	1 and 4; 1980.517 _[7] ***

Notes: Model 1 = four-factor model; Model 2 = three-factor model (GWE and OCBE combined); Model 3 = two-factor model (ECSR, GWE, and OCBE combined); Model 4 = one-factor model (all factors combined). GSV = green shared vision; ECSR = environmental corporate social responsibility; GWE = green work engagement; OCBE = organizational citizenship behavior for the environment; CI = confidence interval; RMSEA = root mean square error of approximation; NFI = normed fit index; CFI = comparative fit index; AIC = Akaike information criterion. $***p < 0.001$

Table 4
Discriminant validity testing.

Variables	1	2	3
1. GSV			
2. ECSR	0.401		
3. GWE	0.241	0.465	
4. OCBE	0.382	0.617	0.422

Notes: Scores below the diagonal represent the Heterotrait-Monotrait ratios between the latent variables. GSV = green shared vision; ECSR = environmental corporate social responsibility; GWE = green work engagement; OCBE = organizational citizenship behavior for the environment.

both serial multiple mediation ($a_1 d_{21} b_2$) and direct effects were found and the direction was positive, ECSR and GWE played *serial multiple and complementary mediation* roles (Zhao et al., 2010).

5. Discussion

By drawing upon the principles of the AMO (Appelbaum et al., 2000) and signaling (Spence, 1973) theories, this paper develops a research framework to explore the direct effect of GSV on OCBE while revealing the single and serial multiple mediation roles of ECSR and GWE, as findings from prior works remained incoherent and under-explored. We examined these interrelationships with data gathered from hotel employees in South Korea, and our research confirms the direct effect of GSV on OCBE in the hospitality industry. It supports the work of Chang (2020), who examined it in the context of environmental maintenance, and Wiraditja et al. (2020), who investigated similar dynamics in the apparel sector. These findings show the generalizability of GSV's positive impact across various industries, highlighting that a shared commitment to sustainability can inspire employees to engage in extra-role behaviors that benefit the organization's environmental objectives (Karatepe et al., 2022).

This study also confirms the significant single mediation impact of ECSR in the relationship between GSV and OCBE. This finding aligns

Table 5

Mean, standardized deviation, and correlation.

Variables	1	2	3	4	5	6	7
1. Gender	1.000						
2. Organizational tenure	0.041	1.000					
3. Hotel grade	−0.011	0.071***	1.000				
4. GSV	−0.015	0.099*	0.157**	1.000			
5. ECSR	−0.006	0.279***	0.277***	0.357***	1.000		
6. GWE	−0.030	0.060	0.139**	0.214***	0.428***	1.000	
7. OCBE	−0.008	0.363***	0.215***	0.331***	0.568***	0.388***	1.000
Mean	0.483	1.715	0.589	4.732	3.335	3.940	3.538
Standard deviation	0.500	1.003	0.493	1.385	0.668	0.899	0.627

Notes: Gender (0 = male, 1 = female) and hotel grade (0 = four-star hotel, 1 = five-star hotel) were operationalized as binary variables. Four categories were used to measure organizational tenure (1 = less than 5 years, 2 = 5–9 years, 3 = 10–14 years, 4 = more than 14 years). GSV = green shared vision; ECSR = environmental corporate social responsibility; GWE = green work engagement; OCBE = organizational citizenship behavior for the environment. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (one-tailed test)

with the research of [Chang et al. \(2020\)](#), who also identified ECSR as a critical factor linking GSV to enhanced environmental performance among employees, which is comparable to employee OCBE in the current work. Thus, organizations that effectively communicate their commitment to environmental responsibility can foster a culture that encourages employees to adopt OCBE as a normative expectation, thereby enhancing organizational sustainability ([Brammer et al., 2022](#)).

Additionally, the single mediation role of GWE in the link between GSV and OCBE is confirmed by the findings. This is consistent with [Karatepe et al.'s \(2022\)](#) research, in which the scholars confirmed the mediation role of GWE in the link between managerial commitment environment, which is comparable to GSV in this study, and employee duty-related and proactive pro-environmental behavior for the organization, which are comparable to OCBE in this study. Employees who perceive a strong GSV increase their engagement in green initiatives, leading to improved OCBE. This suggests that hospitality organizations should invest in initiatives that deliver GSV industry-wide, such as training and development programs focused on long-term sustainability, to realize the full potential of their GSV ([Zhou et al., 2023](#)).

This study also confirms the serial multiple mediation impacts of ECSR and GWE in the association between GSV and OCBE. It supports the work of [Kim et al. \(2023a\)](#), in which scholars confirmed the serial multiple mediation effect of green knowledge sharing (which is comparable to ECSR) and GWE in the link between GHRM (which is comparable to GSV) and employee green behavior beyond that required of them, giving the hotel a competitive advantage due to its environmental responsibility. This finding shows that hotel employees are important in implementing hotels' GSV and ECSR initiatives. It highlights a need for organizations to recognize the interconnectedness of these variables to cultivate a workforce that actively supports environmental initiatives.

5.1. Theoretical contributions

The current research offers several contributions. First, this work stitches together AMO ([Appelbaum et al., 2000](#)) and signaling ([Spence, 1973](#)) theories to develop a novel theoretical framework. This framework develops a comprehensive understanding of how GSV impacts employee voluntary pro-environment behavior in the hospitality context. Unlike many other sectors, where employee motivation may be primarily geared by extrinsic rewards, our findings reveal that hotel employees view GSV as a key indicator of social exchange with management, which is beyond extrinsic or monetary rewards. This alignment not only develops a culture of sustainability but also supports the hospitality industry's unique service-oriented nature, where employee engagement beyond their prescribed roles at work plays an instrumental role for customer satisfaction and brand loyalty in the long run ([Karatepe et al., 2022](#); [Latif et al., 2022](#)). It is particularly important to consider in the hospitality context where employees serve as brand ambassadors for sustainability ([López et al., 2022](#)). The emphasis on

GSV as a predictor of OCBE highlights the importance of aligning organizational goals with employee engagement in environmental initiatives, which remained mixed and fragmented in previous studies.

In this paper, we present how GSV serves as an important signal from top management regarding the commitment to environmental practices. This signaling aspect is particularly important in the hospitality industry, where the direct interaction between employees and customers may exert a profound impact on organizational reputation and sustainability efforts. This paper presents that in the hospitality context, ECSR not only enhances employees' environmental awareness but also develops an emotional connection to the organization's sustainability efforts, motivating employees to engage in voluntary pro-environmental behaviors to boost organizational ethical position ([Pérez et al., 2023](#)). Our findings show that when leadership effectively communicates GSV, it revitalizes the implementation of ECSR initiatives, which, in turn, develops a green work environment among employees. This phenomenon contrasts with other industries, where signals may not be as closely tied to management, employee, and customer interaction or brand perception, emphasizing the hospitality sector's need for a cohesive approach to sustainability that resonates throughout the organization. Unlike industries with more rigid structures and less emphasis on interpersonal relationships, the hospitality sector thrives on teamwork and shared values. Our findings assert that when management and employees receive a strong GSV, they are more likely to internalize these values considering teamwork spirit and shared values, forging a culture of sustainability that is distinctively powerful in hospitality settings ([Brammer et al., 2022](#)). This study identified GSV as the triggering point in AMO ([Appelbaum et al., 2000](#)) and signaling ([Spence, 1973](#)) theories, explaining the interrelationships between GSV, ECSR, GWE, and OCBE, which remained under explored in previous studies.

Our results indicate that ECSR and GWE play instrumental roles in gearing up the momentum of employee green behavior. This insight is key for hospitality managers seeking to enhance employee engagement in sustainability initiatives. Our findings suggest that employees in hospitality are motivated by intrinsic values associated with environmental stewardship and social responsibility, making ECSR and GWE critical mediators in shaping OCBE. This contrasts with other sectors, where financial performance and operational efficiency may overshadow the significance of environmental engagement ([Ren et al., 2021](#)). This research expands on existing literature by investigating the serial multiple mediation impacts of ECSR and GWE in the relationship between GSV and OCBE. A handful of previous studies explored the serial multiple impacts of GWE, green knowledge sharing, and green creativity ([Kim et al., 2023a](#)), and green psychological climate and GWE ([Bhutto et al., 2021](#)). There is no explicit evidence of the serial multiple mediation effects of ECSR and GWE in the existing literature. By demonstrating how ECSR and GWE interact serially to enhance employee green behaviors, this study not only fills a critical gap but also offers a methodological framework for future research. This novel

approach allows for a more nuanced understanding of the factors influencing employee behaviors beyond mere compliance, setting a precedent for subsequent investigations into the intertwined nature of sustainability and organizational behavior in hospitality.

5.2. Practical implications

This work has several practical implications for the hospitality industry. First, this study found that GSV is a crucial driver of OCBE. Although GHRM is an important mechanism to stimulate employee GWE (Çop et al., 2020), hotels need visionary leadership to increase employees' morale so that they remain not only dedicated to performing their assigned duties but also engage in additional green behaviors to achieve hotel pro-environment goals. Hotels can implement an initiative called "Marriott Green Team," encouraging employees from various departments to collaborate on eco-friendly projects, such as reducing plastic usage and conserving water. Regular meetings and updates on progress can help reinforce the shared vision and motivate staff to engage actively in green practices.

Organizational leadership holds the potential to trigger ECSR managers to share organizational sustainability concerns with other employees, which increase employees' morale and shape green attitudes (Zhang et al., 2021). Hotels need to include ECSR concepts in their managerial-level employees onboarding and continuous training programs. For example, hotels could offer workshops on sustainable hospitality practices, covering topics such as energy conservation, waste management, and responsible sourcing. By equipping employees with the knowledge and skills needed to implement sustainable practices, hotels can enhance their staff's capability to contribute to organizational sustainability goals and foster a culture of environmental responsibility. High awareness of green tasks and CSR practices improves OCBE (Chang, 2020).

Hotel management should actively communicate their commitment to sustainability through visible actions and messages. For instance, a hotel can showcase its green initiatives, such as solar panel installations or organic garden projects, through marketing materials and social media campaigns. By consistently signaling their dedication to sustainability, hotel leaders can inspire hotel management and employees to align their behaviors with organizational values, enhancing their engagement and motivation in sustainability efforts. Hyatt Hotels Corporation bolsters their image as visionary leaders in elite hospitality, listening to their colleagues and guests, which has helped Hyatt to attract employees and customers (BMB, 2023). Accor Hotels uses social media and official webpages to promote their commitment to environmental protection across their international chains of hotels, which heightens employee motivation to support the organization's pro-environment principles (Karatepe et al., 2022). Hilton Hotels uses its interactive artificial intelligence reporting and feedback platforms to share Hilton leaders' ecological vision and their commitment to environmentally-friendly corporate values with employees, inducing them to share new green knowledge, new green ideas, engage in GWE, and perform beyond their job responsibilities to improve Hilton's sustainability efforts in offering unique hospitality experience to Hilton customers (Kim et al., 2023a). These practices at brand hotels serve as roadmaps for other hotels to improve their employee morale and receive employee extra green performance beyond pre-defined tasks.

Implementing structured feedback mechanisms, such as suggestion boxes or regular employee forums, allows hotel staff to voice their opinions and ideas regarding sustainability initiatives. For example, a hotel could establish a "Green Innovation Committee" where employees from various departments meet monthly to discuss sustainability challenges and propose solutions. This inclusive approach empowers employees and develops a sense of ownership and accountability for the hotel's sustainability efforts, fueling the commitment to ECSR. By sharing new green visionary ideas, adopting ECSR practices, engaging in green work, and motivating fellow employees to perform extra, hotels

may, for example, publicize the most exemplary green staff each month on their official webpages and social media. Such a practice sends strong CSR signal and may motivate employees to voluntarily engage in additional pro-environment actions (Kim et al., 2023a). Without GSV, hotels may find it difficult to stimulate employee commitment to follow hotel ECSR initiatives and voluntarily contribute to the realization of hotel green goals.

5.3. Limitations and follow-up studies

Limitations associated with the current research are noted, and their important future research avenues are provided. From the perspective of theory and literature, first, future research may benefit from incorporating national culture and leadership styles into our proposed research model to garner additional insights.

Second, this work examined the effect of two mediators (ECSR and GWE). Follow-up research should explore the single and serial multiple mediation impacts of green organizational support, green organizational identification (Aboramadan, 2022), and green knowledge sharing (Karatepe et al., 2022; Kim et al., 2023a) alongside ECSR and GWE in the association between GSV and OCBE to glean a better understanding of green hospitality management. Additionally, the conceptual understanding of the role of GSV is in a state of flux (compare findings of this study with Latif et al., 2022), which demands future research from different perspectives in order to reach a conclusive understanding of GSV in determining OCBE.

Third, from the perspective of methodology, the findings of this research are based on data collected from upper-upscale and luxury hotels in Seoul, South Korea. We encourage researchers to investigate whether the findings of this study are consistent across different countries and industries, such as airlines, restaurants, and healthcare.

Fourth, according to scholars who have extensively studied CMB (e.g., Baumgartner and Steenkamp, 2001; Doty and Glick, 1998; Park et al., 2007; Podsakoff et al., 2003, 2024; Williams and Anderson, 1994; Williams et al., 2010), several major causes of CMB can be effectively mitigated by measuring predictor and criterion variables from different sources (i.e., ensuring response sources are independent) and incorporating temporal separation. The use of different response sources helps prevent shared variance due to the measurement method (Conway, 2002). This separation reduces potential recall cues or the respondents' consistency bias (i.e., consistency effect) (cf. Johns, 1994; Podsakoff and Organ, 1986). As a result, CMV will likely be significantly reduced. However, in this study, due to the characteristics of the variables, survey conditions, and other research constraints, it was not possible to separate the response source and the timing of data collection between the predictor and criterion variables. Although only 6.5 % of the observed constructs were explained by the common method factor, the relationship between GSV and GWE was found to be distorted. Therefore, as Podsakoff et al. (2024) strongly recommend, future studies should ensure that the response source and timing of data collection are separated between predictor and criterion variables to prevent potential distortions in the relationships between them.

Finally, Harman's single-factor technique, which utilizes both exploratory and confirmatory factor analysis approaches, does not identify specific sources of method variance that affect the relationships of interest. It assumes that a single source of method variance influences all items, which is problematic because multiple method variance sources are likely to exist in any survey study (cf. Podsakoff et al., 2024). Specifically, Harman's technique assumes that CMV can only be detected when a single factor explains more than 50 % of the variance in the ratings of substantive constructs. The arbitrary nature of this 50 % threshold makes it a questionable criterion (Podsakoff et al., 2003). Consequently, Harman's single-factor technique is a conservative method with low statistical power (Baumgartner et al., 2021; Schwarz et al., 2017; Podsakoff et al., 2024). Despite its limitations, it remains widely used in various fields, including hospitality and tourism (e.g.,

Bhutto et al., 2021; Gip et al., 2024; Olorunsola et al., 2024; Pham et al., 2020; Serrano-Archimi et al., 2018; Shah et al., 2020; Xin and Wang, 2023; Zhang et al., 2021; Zhao et al., 2024). Given its low statistical power and several drawbacks, Harman's single-factor technique cannot adequately control for CMB. Therefore, in line with the recommendations of Podsakoff et al. (2024), we strongly advise against using this technique in future studies.

CRediT authorship contribution statement

Kim Taegoo Terry: Writing – review & editing, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kim Woo Gon:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology. **Majeed Salman:** Writing – review & editing, Writing – original draft, Project administration, Methodology. **Roh Hanna:** Visualization, Resources, Investigation, Data curation.

Declaration of Competing Interest

NONE. This manuscript has not been published or presented elsewhere in part or in entirety and is not under consideration by another journal. All study participants provided informed consent, and the study design was approved by the appropriate ethics review board. We have read and understood your journal's policies, and we believe that neither the manuscript nor the study violates any of these. There are no conflicts of interest to declare.

Appendix A. Measurement items and their sources

Green shared vision (Chen et al., 2015)

- “GSV1. A commonality of environmental goals exists in my hotel.”
- “GSV2. A total agreement on the strategic environmental direction of the organization exists in my hotel.”
- “GSV3. All members in the organization are committed to the environmental strategies.”
- “GSV4. Employees of the organization are enthusiastic about the collective environmental mission of the organization.”

Environmental corporate social responsibility (Park and Levy, 2014; Türker, 2009)

- “ECSR1. My hotel incorporates environmental concerns into decision processes.”
- “ECSR2. My hotel participates in activities which aim to protect and improve natural environment.”
- “ECSR3. My hotel provides products and services to customers which minimize environmental impacts.”
- “ECSR4. My hotel implements special programs to decrease its negative impact on the environment.”

Green work engagement (Aboramadan, 2022)

- “GWE1. My environmental-related tasks inspire me.”
- “GWE2. I am proud of the environmental work that I do.”
- “GWE3. I am immersed in my environmental work.”
- “GWE4. I am enthusiastic about my environmental tasks at my job.”
- “GWE5. I feel happy when I am working intensely on environmental tasks.”
- “GWE6. With environmental tasks at my job, I feel bursting with energy.”

Organizational citizenship behavior for the environment (Boiral and Paille, 2012)

• OCBE1. Eco-initiatives

“ECO11. In my work, I weigh the consequences of my actions before doing something that could affect the environment.”

“ECO12. I voluntarily carry out environmental actions and initiatives in my daily work activities.”

“ECO13. I make suggestions to my colleagues about ways to protect the environment more effectively, even when it is not my direct responsibility.”

• OCBE2. Eco-civic engagement

“ECOCE1. I actively participate in environmental events organized in and/or by my hotel.”

“ECOCE2. I stay informed of my hotel's environmental initiatives.”

“ECOCE3. I undertake environmental actions that contribute positively to the image of my organization.”

“ECOCE4. I volunteer for projects, endeavors or events that address environmental issues in my organization.”

• OCBE3. Eco-helping

“ECO11. I spontaneously give my time to help my colleagues take the environment into account in everything they do at work.”

“ECO12. I encourage my colleagues to adopt more environmentally conscious behavior.”

“ECO13. I encourage my colleagues to express their ideas and opinions on environmental issues.”

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